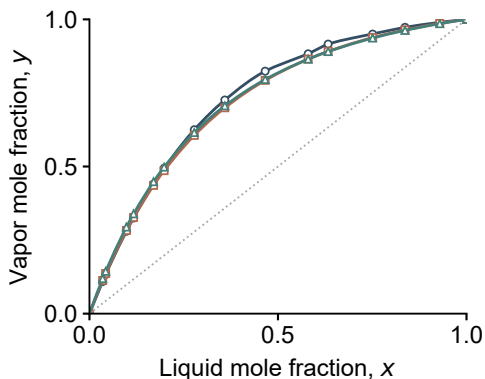


n-Heptane / *n*-nonane: equilibrium and distillation

$P = 101.3 \text{ kPa}$ | $q = 1$ | $z_F = 0.466$ | $x_D = 0.995$ | Recovery = 98%

a VLE reference and predictions



b VLE prediction errors

Metric	ThermoFormer	UNIFAC
Vapor-composition MAE	0.010465	0.011976
Vapor-composition RMSE	0.014384	0.014328
Maximum composition error	0.031195	0.028319
Temperature MAE (K)	0.6593	0.5157
Temperature RMSE (K)	0.9204	0.6221
Composition MAE, mixtures	0.011960	0.013687
Temperature MAE, mixtures (K)	0.7432	0.5719

Rows 1-5: all 16 states; rows 6-7: 14 mixture states.
Composition errors are dimensionless.

c Common material balance

Stream	Total flow	x_{C7}	C_7 flow	C_9 flow
Feed	1.000000	0.466000	0.466000	0.534000
Distillate	0.458975	0.995000	0.456680	0.002295
Bottoms	0.541025	0.017227	0.009320	0.531705

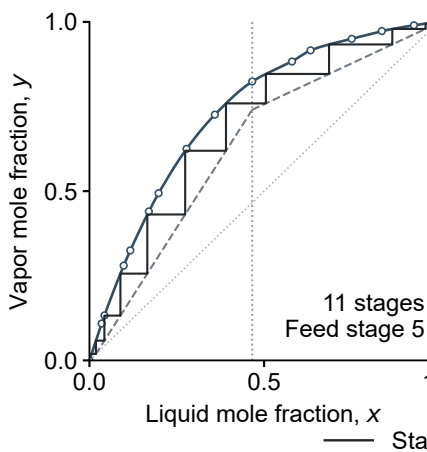
Flows: mol s⁻¹; C_7 : *n*-heptane; C_9 : *n*-nonane.

d Design and interpolation sensitivity

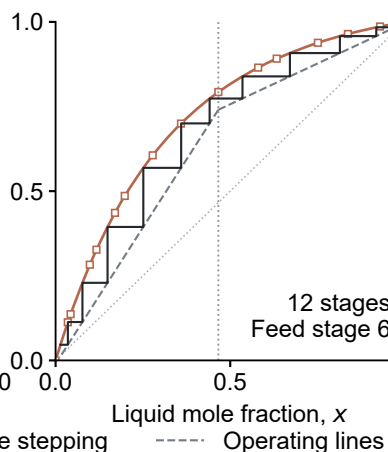
Source	Interp.	R_{min}	N	Feed
ThermoFormer	PCHIP	0.618701	12	6
ThermoFormer	Linear	0.618701	13	6
Experiment	PCHIP	0.477654	11	5
Experiment	Linear	0.477654	11	5
UNIFAC	PCHIP	0.604579	12	6
UNIFAC	Linear	0.604579	12	6

Common reflux ratio: $R = 0.928052$ | N includes the equilibrium reboiler and excludes the total condenser.

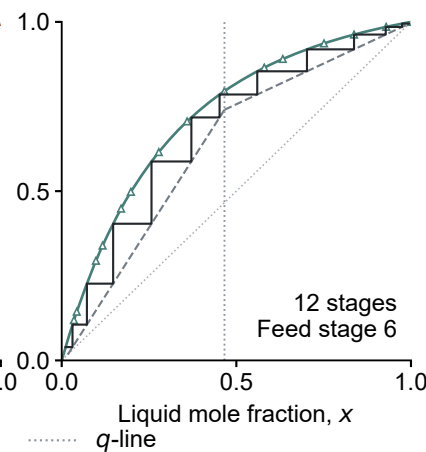
e Experiment



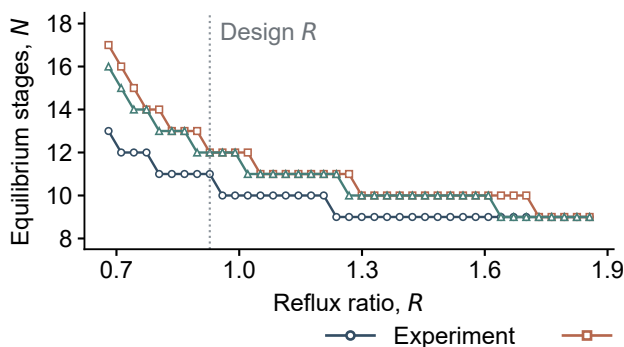
f ThermoFormer



g UNIFAC



h Reflux sensitivity: PCHIP



i Reflux sensitivity: linear

